Chapter 3, Section 12

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1. Let Y be a scheme of finite type over an algebraically closed field k. Show that the function

$$\varphi(y) = \dim_k(\mathfrak{m}_y/\mathfrak{m}_y^2)$$

is upper semicontinuous on the set of closed points of Y.

Proof. Apply (12.7.2) to the sheaf of relative differentials $\Omega_{Y/k}$. For each closed point $y \in Y$, we have a natural isomorphism (II, 8.7A)

$$\mathfrak{m}_y/\mathfrak{m}_y^2 \cong \Omega_{Y/k} \otimes k(y).$$

□

2. Let $\{X_t\}$ be a family of hypersurfaces of the same degree in $\mathbb{P}_k^n$. Show that for each i, the function $h^i(X_t, \mathcal{O}_{X_t})$ is a constant function of t.

Proof. Hypersurfaces of same degree in projective space $\mathbb{P}^n$ have the same cohomology. For any hypersurface $X \subset \mathbb{P}^n$ of degree d, we have an exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_X \longrightarrow 0,$$

and $h^i(X, \mathcal{O}_X)$ is independent of the choice of X. □

4. Let Y be an integral scheme of finite type over an algebraically closed field k. Let $f : X \to Y$ be a flat projective morphism whose fibers are all integral schemes. Let $\mathcal{L}, \mathcal{M}$ be invertible sheaves on X, and assume for each $y \in Y$ that $\mathcal{L}_y \cong \mathcal{M}_y$ on the fiber X_y. Then show that there is an invertible sheaf $\mathcal{N}$ on Y such that $\mathcal{L} \cong \mathcal{M} \otimes f^*\mathcal{N}$.

Proof. If f is flat, then $\mathcal{L} \otimes \mathcal{M}^{-1}$ is flat over Y. For each $y \in Y$, $(\mathcal{L} \otimes \mathcal{M}^{-1})_y \cong \mathcal{O}_{X_y}$ on the fibre X_y, and each X_y is integral and projective over $k(y)$, so we have

$$h^0(X_y, (\mathcal{L} \otimes \mathcal{M}^{-1})_y) = h^0(X_y, \mathcal{O}_{X_y}) = 1.$$

Thus, $f_*(\mathcal{L} \otimes \mathcal{M}^{-1})$ is locally free of rank one on Y (12.9), say $f_*(\mathcal{L} \otimes \mathcal{M}^{-1}) = \mathcal{N}$ for some $\mathcal{N} \in \operatorname{Pic} Y$. Applying f^* to both sides, and using the fact that the natural morphism

$$f^* f_*(\mathcal{L} \otimes \mathcal{M}^{-1}) \to \mathcal{L} \otimes \mathcal{M}^{-1}$$

is actually an isomorphism because $f_*(\mathcal{L} \otimes \mathcal{M}^{-1})$ is locally free of rank one, we obtain our desired isomorphism. □

5. Let Y be an integral scheme of finite type over an algebraically closed field k. Let $\mathcal{E}$ be a locally free sheaf on Y, and let $X = \mathbb{P}(\mathcal{E})$. Then show that $\operatorname{Pic} X \cong (\operatorname{Pic} Y) \times \mathbb{Z}$. This strengthens (II, Ex. 7.9).

Proof. Let r be the rank of $\mathcal{E}$, $\mathcal{O}_X(1)$ the canonical invertible sheaf of X, and $\pi : X \to Y$ the natural projection morphism. For any point $y \in Y$, let $k(y)$ be the residue field of y, and let $\mathbb{P}_y$ be the fiber of π over y, which is a projective r-space over $k(y)$. There is an obvious map

$$\begin{aligned} \varphi : \operatorname{Pic} Y \times \mathbb{Z} &\to \operatorname{Pic} X \\ (\mathcal{L}, d) &\mapsto \mathcal{O}_X(d) \otimes \pi^*\mathcal{L}. \end{aligned}$$

First, we show φ is injective. If we have an isomorphism

$$\mathcal{O}_X(d) \otimes \pi^*\mathcal{L} \cong \mathcal{O}_X(d') \otimes \pi^*\mathcal{L}',$$

then it is clear by checking on fibres that $d = d'$, so it remains to show $\pi^* : \operatorname{Pic} Y \to \operatorname{Pic} X$ is injective. We have $\pi_*\mathcal{O}_X = \mathcal{O}_Y$ (II, 7.11a), and for any $\mathcal{N} \in \operatorname{Pic} Y$, the projection formula gives

$$\pi_*\pi^*\mathcal{N} \cong \pi_*\mathcal{O}_X \otimes \mathcal{N} \cong \mathcal{N},$$

so if $\pi^*\mathcal{N} \cong \mathcal{O}_X$, then $\mathcal{N} \cong \pi_*\mathcal{O}_X \cong \mathcal{O}_Y$. Hence, φ is injective.

The hard part is to show φ is surjective. Since π is a flat projective morphism, we can apply the results of §12. In particular, if $\mathcal{M}$ is an invertible sheaf on X, then $\mathcal{M}_y$ is an invertible sheaf on $\mathbb{P}_y$ for every $y \in Y$, so we can write $\mathcal{M}_y \cong \mathcal{O}(d_y)$ for some $d_y \in \mathbb{Z}$. If we can show d_y is independent of $y \in Y$, then by (Ex. 12.4) there exists an invertible sheaf $\mathcal{N}$ on Y such that $\mathcal{M} \cong \mathcal{O}_X(d) \otimes \pi^*\mathcal{N}$ for some $d \in \mathbb{Z}$, which proves φ is surjective.

We proceed by showing the minimum value of d_y is obtained when $y = \eta$ is the generic point of Y. Suppose there exists $y' \in Y$ such that $d_{y'} < d_\eta$. After twisting by $\mathcal{O}_X(-d_{y'})$, we can assume $d_{y'} = 0$. By (12.8), the function

$$h^0(y, \mathcal{M}) = \dim_{k(y)} \mathrm{H}^0(\mathbb{P}_y, \mathcal{O}_{\mathbb{P}_y}(d_y))$$

is upper semicontinuous on Y (12.8), and we have the strict inequality

$$\dim_{k(y')} \mathrm{H}^0(\mathbb{P}_{y'}, \mathcal{O}_{\mathbb{P}_{y'}}) < \dim_{k(\eta)} \mathrm{H}^0(\mathbb{P}_\eta, \mathcal{O}_{\mathbb{P}_\eta}(d_\eta))$$

by the explicit calculations of (5.1). But this implies the set $\{y \in Y \mid h^0(y, \mathcal{M}) \geq h^0(\eta, \mathcal{M})\}$ is a closed set containing η and not containing y', a contradiction. Hence, we have $d_\eta \leq d_y$ for all $y \in Y$.

Next, we prove the inequality in the opposite direction. By Serre duality (5.1) (7.7), we have

$$\begin{aligned} h^{r-1}(y, \mathcal{M}) &= \dim_{k(y)} \mathrm{H}^{r-1}(\mathbb{P}_y, \mathcal{O}_{\mathbb{P}_y}(d_y)) \\ &= \dim_{k(y)} \mathrm{H}^{r-1}(\mathbb{P}_y, \mathcal{O}_{\mathbb{P}_y}(-d_y - r)) \\ &= h^0(y, \mathcal{O}_X(-r) \otimes \mathcal{M}^{-1}), \end{aligned}$$

and $h^{r-1}(y, \mathcal{M})$ is also an upper semicontinuous on Y. Suppose there exists $y' \in Y$ such that $d_\eta < d_{y'}$. After twisting by $\mathcal{O}_X(-d_\eta + r)$, we can assume $d_\eta = r$. In particular $d_{y'}$ is a positive integer. Again, we have the strict inequality

$$\dim_{k(y')} \mathrm{H}^0(\mathbb{P}_{y'}, \mathcal{O}_{\mathbb{P}_{y'}}(-d_{y'} - r)) < \dim_{k(\eta)} \mathrm{H}^0(\mathbb{P}_\eta, \mathcal{O}_{\mathbb{P}_\eta}),$$

arriving at the same contradiction as above. Hence, d_y is independent of $y \in Y$. □